GGG–H–142b
OCTOBER 6, 1953
SUPERSEDING
Int. Fed. Spec. GGG–H–00142a (Navy Ships)
29 October 1952

FEDERAL SPECIFICATION

HELMET, CONSTRUCTION WORKER'S

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers safety helmets for protection of the wearer's head against impacts of objects and contacts with exposed electrical conductors. The helmets are intended for protection against these hazards in mining, building construction, tunnelling, timber work, and manufacturing.

1.2 Classification.

1.2.1 *Types.*—The helmets shall be furnished in the following types, as specified:

Type I.—Full brim.

Type II.—Brimless with visor.

1.2.2 *Sizes.*—The helmets shall be furnished in the head sizes, as specified (see 3.4.2 and 6.1).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standard, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

L–P–406—Plastics, Organic: General Specifications, Test Methods.

CCC–C–418—Cloth, Coated, Upholstery (Vinyl Resin Base).

Military Standard:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Federal Specifications and the Index of Federal Specifications and Standards may be obtained upon application accompanied by check, money order, cash, or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, SW., Washington 25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Prices may be obtained from the Index of Federal Specifications and Standards or from the GSA Regional Offices. Single copies of this specification and other product specifications required for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Specifications and standards applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Qualification.—Safety helmets furnished under this specification shall be a product which has been tested and has passed the qualification tests specified in section 4 (see 6.2).

3.2 Component parts.—Each helmet shall consist of a shell, headband, crown straps forming a cradle or hammock for supporting the helmet on the wearer's head, and if specified in the contract or order an adjustable chinstrap and a winter lining.

3.3 Material.—All materials coming into contact with the skin shall be nonirritating.

3.3.1 *Shell.*—The shell shall be of smooth, hard material which shall conform to the requirements specified in this specification.

3.3.2 *Headband.*—Unless otherwise specified, the headband shall be genuine vegetable-tanned leather, full grained and soft. The headband shall be smoothly finished on the surface which will be in contant with the head. If specified in the contract or order, headband may be made of artificial leather conforming to type II, class 3 of Federal Specification CCC–C–418.

3.3.3 *Crown straps.*—Crown straps shall be of webbing, closely woven.

3.3.4 *Accessory materials.*—If chinstraps and winter linings are specified in the contract or order, they shall be made of the following materials:

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3.3.4.1 *Chinstrap.*—The adjustable chinstrap shall be closely woven cotton or linen webbing or of genuine leather.

3.3.4.2 *Winter lining.*—The winter lining, shall consist of two layers of closely woven fast-dyed fabric. The outer layer shall be a water-repellent material and the inside layer, non-water-repellent, plain-woven cotton flannel with the nap on the inside surface.

3.4 Design and construction.

3.4.1 *Shell.*—The shell shall be of one-piece, seamless construction, dome shaped, with a full brim for type I helmets and with a visor for type II helmets.

3.4.1.1 *Dome.*—The dome shall be designed to provide the inside clearances as specified in 3.4.3.

3.4.1.2 *Brim, type I.*—The brim for type I helmets shall be an integral extension of the shell dome, and shall be continuous around the helmets. The width of the brim shall be not less than 1½ inches and not more than 3 inches measured from the inside edge of the shell, except that the width at the front may taper to not less than 1¼ inches. With the helmet held in a horizontal position, the brim shall slope downward in a straight line at an angle of not less than 15° nor more than 37° from the horizontal.

3.4.1.3 *Visor, type II.*—The visor for type II helmets shall extend forward from the crown not less than 1½ inches, and shall be not less than 5½ inches wide at the peak.

3.4.2 *Headbands.*—The headbands shall be adjustable or nonadjustable, as specified (see 6.1). If adjustable, they shall be suitable for the range of head sizes specified (see 1.2.2. and 6.1). If nonadjustable, they shall be furnished in the specified head sizes. The headband shall conform in size to the marking of the helmet or shall permit adjustment to the marked size or sizes. If made of leather, the thickness, determined at 10 points uniformly spaced throughout the circumference of the headband, shall be not less than 1/32 inch, when measured with a gage having a 5/16-inch-diameter contact foot under a 1-pound load. The surface of the headband in contact with the head shall be not less than 1¼ inches in width. The headband shall be securely attached either to the helmet shell or to the crown straps for assuring its proper position in the helmet. The manner of the attachment should not interfere with the adjustments of the crown straps. Padding or stiffener strips, if used as a regular part of the headband, shall be secured to the leather.

3.4.3 *Crown straps.*—Crown straps shall be not less than ½ inch in width. These straps when adjustably and properly interlaced shall form a cradle or hammock for supporting the helmet on the wearer's head, for holding the helmet away from the head and thus acting as a shock absorber when the helmet is struck. The number of straps shall be adequate and their location in the helmet shall be suitable for the above purpose. The straps may have one end attached with the loose end having a sewn loop, or may have both ends attached forming a bent loop. Through these loops a strong cord or lace shall pass for adjusting the cradle to suit the wearer's head, leaving a clearance of not less than 1½ inches between the top of the wearer's head and the underside of the top of the crown. The cord or lace shall provide for adjustment of clearance between the top of the wearer's head and the underside of the top of the crown up to at least 1½ inches.

3.4.4 *Accessories.*—If specified in the contract or order, the following accessories shall be furnished with the helmet:

3.4.4.1 *Chinstrap.*—The adjustable chinstrap shall be of not more than two pieces of webbing or leather of not less than ½ inch in width and not less than 16 inches in adjusted length excluding the attachments to the helmet. The means for adjusting the chinstrap shall assure a secure hold of the helmet on the wearer's head and quick removal of the helmet by releasing the strap. The chinstrap shall also be adjustable for wearing at the back of the head. The webbing shall have no frayed or loose edges that may unravel. The leather, if used in the chinstrap, shall be suitable thickness, full grained, and smoothly finished on the surface that will be in contact with the chin. All metal parts, including rivets, shall be smooth and free from sharp or rough edges or projections. Rivet heads shall be round.

3.4.4.2 *Winter lining.*—The winter lining shall be designed to cover the skull, neck, and

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ears. The skull cap, part of the winter lining, shall be attached to the helmet cradle or to the helmet shell in such a manner that it can be readily removed and reattached. The earlug and neck part may be made either in one piece with the skull cap or attachable to it by tabs or a zipper. The neck and earlug shall be drawn close to the face by a lace or a chin strap.

3.5 Physical requirements.

3.5.1 *Insulation resistance.*—The helmet shall not show a leakage current in excess of 3 milliamperes when tested as specified in 4.5.2.

3.5.2 *Impact resistance.*—When helmets are tested in accordance with 4.5.3, the average diameter of the impressions shall not exceed 4.5 millimeters. No individual specimen shall show a diameter of impression greater than 4.9 millimeters. (The diameters of impressions specified correspond, when using 2S–O aluminum, to transmitted forces of approximately 850 and 1,000 pounds, respectively.)

3.5.3 *Penetration strength.*—The helmet shall be neither dented nor pierced for more than ⅜ inch, nor shall the helmet be forced down over the standard head form, nor shall the straps pull out or break when tested as specified in 4.5.4.

3.5.4 *Edge strength, type II.*—When tested in accordance with 4.5.5, type II helmets shall show a deflection under a 40-pound load of not more than ¾ inch and an ultimate strength of not less than 50 pounds.

3.5.5 *Flammability.*—The thinnest section of the shell shall not burn at a rate greater than 3 inches per minute when tested as specified in 4.5.6.

3.5.6 *Water absorption.*—The helmet shell material shall absorb not more than 5 percent of water when tested as specified in 4.5.7.

3.5.7 *Weight.*—The weight of each complete helmet without winter lining and chinstrap shall not exceed 14½ ounces.

3.5.8 *Ventilation.*—Provision shall be made between the shell and the headband, or in the headband proper, for a free passage of air for ventilation purposes.

3.6 Interchangeability.—The headband and crown-strap assembly shall be easily removable and replaceable, and shall be interchangeable among shells of the same size furnished by the same manufacturer, provided that a single type of lining is furnished. Where two types of linings are involved neither lining need be usable as a replacement in shells originally provided with the other type.

3.7 Instruction sheet.—An instruction sheet which explains the proper method of wearing and adjusting the helmet shall be furnished with each helmet. The sheet shall be stowed in the helmet so as to remain where placed until intentionally removed.

3.8 Marking.

3.8.1 *Shell.*—Each helmet shell shall be marked in a permanently legible manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined.

3.8.2 *Headband.*—The head size or range of sizes to which the headband is adjustable shall be marked on labels cemented to, or embossed upon, the helmet shell and sewn or cemented to the replaceable crown-strap assembly (see 3.4.2).

3.9 Workmanship.—The workmanship shall be first class in every respect. The safety helmets shall have no sharp edges or burrs, and shall be free from any other defects which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Qualification tests at a Government laboratory.—Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.5.

4.2 Sampling for lot acceptance.

4.2.1 *Lot.*—All helmets of one type presented for delivery at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.2.2 *Sampling for inspection.*—A random sample of helmets shall be selected as specified in table I from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the following sampling inspection requirements in accordance with Military Standard MIL–STD–105.

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TABLE I.—*Sampling for visual and dimensional inspection AQL (approx.)=2.5 percent defective*

Number of helmets in inspection lot	Number of helmets in sample	Acceptance number (defectives)	Rejection number (defectives)
40 or under	10	0	1
41 to 110	15	1	2
111 to 300	25	1	2
301 to 500	35	2	3
501 to 800	50	3	4
801 to 1,300	75	4	5
1,301 to 3,200	110	6	7
3,201 and over	150	8	9

4.2.3 *Sampling for lot-acceptance tests.*—Samples shall be selected at random from each inspection lot by the Government inspector for lot-acceptance tests in accordance with table II.

TABLE II.—*Sampling for lot-acceptance tests*

Number of helmets in lot	Number of helmets in sample
16 to 40	3
41 to 110	6
111 to 300	9
301 to 500	12
501 to 800	15
801 to 1,300	18
1,301 to 3,200	24
3,201 and over	36

4.3 Inspection and tests.

4.3.1 *Visual and dimensional inspection.*—Each of the sample helmets selected in accordance with table I shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Any helmet in the sample containing one or more visual or dimensional defects shall be rejected, and, if the number of defective helmets in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.2 *Lot-acceptance tests.*—Samples selected in accordance with 4.3.2 shall be subjected to the tests specified in 4.5.1 through 4.5.5.4 except that the impact test of 4.5.3 shall be conducted at the prevailing atmospheric temperature in lieu of the temperatures specified in 4.5.3.3. If any sample helmet fails one or more of these tests, the entire lot shall be rejected.

4.4 Rejected lots.—Rejected lots may be offered again for Government inspection provided the contractor has removed all nonconforming helmets. The Government inspector shall again select and examine samples from such resubmitted lots to verify compliance with this specification.

4.5 Test procedures.

4.5.1 *Preparation of samples.*—The sample helmets shall be selected as specified in table II. Where it is evident that the sample helmets under test have a protective coating over the basic material, the entire exterior surface of the shell shall be abraded until the basic material is exposed using No. 60-grit Garnet paper. All samples shall then be conditioned in an oven for 96 hours at 50°±2° C., then cooled in a desiccator and exposed for 96 hours in an atmosphere having 50±2 percent relative humidity and a temperature of 25°±2° C. Immediately following conditioning, ⅓ of the samples shall be subjected to the test specified in 4.5.3, ⅓ to the test specified in 4.5 4, and the remainder to the test specified in 4.5.5.

4.5.2 *Insulation-resistance test.*

4.5.2.1 *Test specimens.*—All of the helmets in one of the groups specified in 4.5.1 prior to subjection to the impact test or to the penetration test shall be subjected to the insulation-resistance test.

4.5.2.2 *Apparatus.*—The test apparatus shall consist of the following:

(*a*) A vessel containing a solution of 1 ounce of sodium chloride per gallon of water and of sufficient size to submerge an inverted helmet to within ½ inch of the junction of brim with the dome.

(*b*) Wire frame for suspending inverted helmet in sodium-chloride solution.

(*c*) Source of 60-cycle alternating current with 2,200 volts (r. m. s.).

(*d*) Wiring and terminals for application of voltage across dome of helmet.

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(*e*) Voltmeter of sufficient capacity.

(*f*) Milliammeter of sufficient capacity and accuracy to measure the specified current.

4.5.2.3 *Procedure.*—An inverted helmet with the head-harness assembly removed shall be placed on the wire frame and submerged in the solution to within ½ inch of the junction of brim with the dome. The helmet shall also contain the solution to the same level. If the helmet is provided with holes in the crown near the brim, the helmet shall be submerged to within ½ inch of the holes. One terminal from the current source shall be connected to the wire frame which supports the helmet and the other shall be immersed in the sodium-chloride solution which is contained by the helmet. The 2,200 volts shall then be applied for 1 minute. If passage of current is evidenced by deflection of the voltmeter, the helmet shall be considered as having failed to conform to 3.5.1. If no deflection of voltmeter occurs, a milliammeter is placed in the circuit and a reading taken.

4.5.2.4 *Report.*—For each specimen, the leakage current or evidence of complete breakdown shall be reported.

4.5.3 *Impact test.*

4.5.3.1 *Test specimens.*—All helmets in the first group prepared in accordance with 4.5.1 shall be subjected to the impact test.

4.5.3.2 *Apparatus.*—The test apparatus shall consist of the following:

(*a*) A standard head form (see 6.3) of the same head size as the helmet under test. The standard head forms are of 4 sizes: small, medium, large and extra large, representing head sizes of 6¾, 7, 7¼, and 7½, respectively. Test-block forms which simulate the standard head forms may be employed. The blocks shall duplicate the crown dimensions and contours from the top down to the ear and eye levels, including the top portion of the ears. Each test block shall have a hole drilled vertically from the top of the crown to facilitate measurement of hat crown clearance.

(*b*) A Brinell-hardness penetrator assembly as shown on figure 1. The impression bar shall be of 2S–O aluminum not less than ¼ inch thick and having a Brinell hardness of 20 to 25 as measured with a 500-kilogram load using a 10-millimeter ball. The Brinell penetrator shall be a hardened-steel ball ½ inch in diameter.

(*c*) An 8-pound spherical weight, approximately 3.8 inches in diameter.

(*d*) A Brinell microscope or other micrometer microscope accurate to 0.1 millimeter.

4.5.3.3 *Test conditions.*—Impact tests shall be conducted with the specimens maintained at temperatures of 20°±2° F. and 120°±2° F. Specimens shall be subjected to the test temperature for at least 4 hours prior to the impact test.

4.5.3.4 *Procedure.*—The specimens specified in 4.5.3.1 shall be mounted on the appropriate standard head form in a position similar to that when worn on a man's head. For purposes of testing, headbands of sizes intermediate to the standard form size shall be padded with felt to reduce the headband to fit the nearest applicable form size; for example, a size 7⅛ helmet shall be padded to fit a size 7 form. The helmets shall be adjusted to a clearance of 1¼ inches between crown of test block and the underside of the top of helmet crown by placing a 25-pound weight on the helmet and measuring by means of a rod inserted in the hole drilled in test block (see 4.5.3.2 (a)). The helmet-and-block assembly shall then be mounted on the Brinell-hardness penetrator assembly, figure 1. The 8-pound spherical weight shall then be dropped vertically on the center of the crown from a height of 5 feet, measured from the bottom of the ball to the top of the helmet. The diameter of the impression, produced on the impression plate as a result of the impact, is then measured to the nearest 0.1 millimeter with the Brinell or other suitable micro-meter microscope.

4.5.3.5 *Report.*—The average diameter of impression of all helmets tested under one group and the greatest individual diameter of impression from the same group shall be reported.

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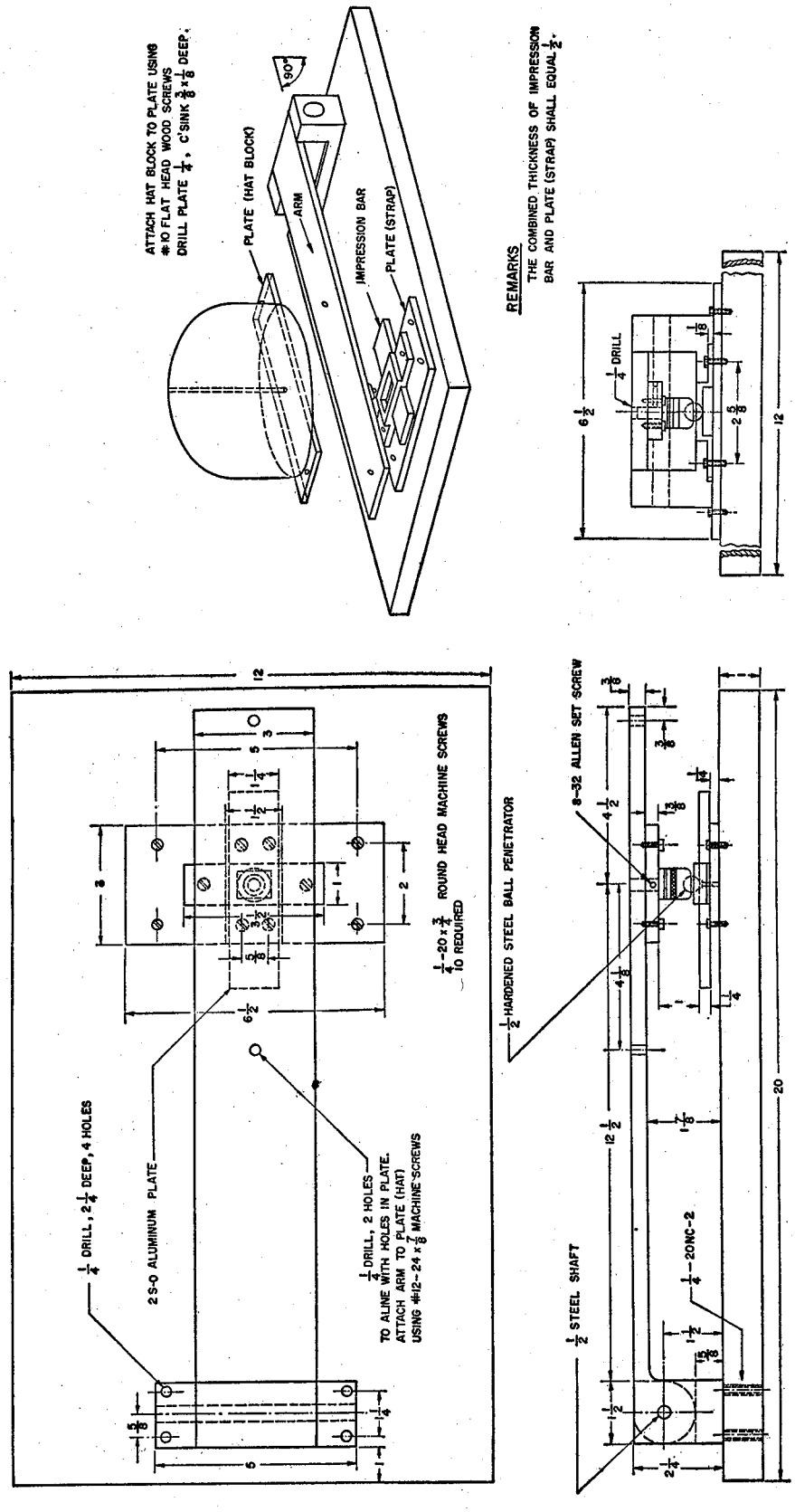


FIGURE 1.—Brinell-hardness penetrator assembly.

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4.5.4 *Penetration test.*

4.5.4.1 *Test specimens.*—All helmets in the second sample group prepared in accordance with 4.5.1 shall be subjected to the penetration test.

4.5.4.2 *Apparatus.*—The test apparatus shall consist of standard head forms or test blocks as specified in 4.5.3.2 (*a*) and a 16-ounce plum bob with a steel point having an included angle of 36°.

4.5.4.3 *Procedure.*—The specimen shall be mounted on a test block as specified in 4.5.3.4 and the plumb bob dropped squarely onto the center of the crown from the height of 10 feet, measured from the bottom point of the plumb bob to the top of the helmet. The depth the plumb bob has pierced the helmet shall be measured to the nearest 1/32 inch and shall include the thickness of the shell material.

4.5.4.4 *Report.*—For each specimen, the depth of penetration to the nearest 1/32 inch shall be reported.

4.5.5 *Edge strength test, type II.*

4.5.5.1 *Test specimens.*—All helmets in the third group prepared in accordance with 4.5.1 shall be subjected to the edge strength test.

4.5.5.2 *Apparatus.*—The test apparatus shall consist of the following:

(*a*) A universal testing machine capable of applying the loading specified in 3.5.4 at a rate of head travel of 0.05±0.01 inch per minute.

(*b*) A dial gage for measuring deflections under loading and accurate to 0.01 inch.

(*c*) A jig designed to support a type II safety helmet on its side on the bedplate of the testing machine so that a plane intersecting the edge of the crown is vertical and the long axis of the helmet horizontal.

4.5.5.3 *Procedure.*—With a specimen mounted in the jig, the traveling head of the test machine shall be brought into contact with the edge of crown and the rate of deflection adjusted to 0.05±0.01 inch per minute. The dial gage shall be adjusted to measure the travel of the moving head. Deflections shall be measured at intervals of loading of 10 pounds. Loading shall be continued to failure.

4.5.5.4 *Report.*—For each specimen, the deflection under a load of 40 pounds and the ultimate load at failure shall be reported.

4.5.6 *Flammability test.*

4.5.6.1 *Test specimens.*—All helmets in one of the three groups prepared in accordance with 4.5.1, and after subjection to one of the strength tests specified in 4.5.3, 4.5.4 or 4.5.5, shall be subjected to the flammability test.

4.5.6.2 *Apparatus.*—The test apparatus shall consist of the following:

(*a*) Rigid support and clamps for mounting test strips in specified position.

(*b*) Alcohol lamp or gas burner.

(*c*) Stop watch.

4.5.6.3 *Procedure.*—Three strips approximately 1/2 inch wide and 5 inches long shall be cut from the brim or sides of the crown, whichever is thinner. These strips shall be cut from a portion of the hat so as to give as flat a section as practicable. Lines shall be drawn across each strip at 1/2-inch intervals, measured along the surface and starting from one end. Clamp the other end to the rigid support so that the longitudinal axis is horizontal and the transverse axis is inclined 45°. The alcohol lamp or gas burner, with a neutral, blue flame 1/2 to 3/4 inch in height, shall be placed under the free end of the strips, so adjusted that the flame tip is just in contact with the material. A stop watch shall be started when the flame is placed in contact with the strip under test. At the end of 30 seconds the flame shall be removed and the strip allowed to burn noting the rate of burning.

4.5.6.4 *Report.*—The rate of burning shall be reported as the average for the three specimens.

4.5.7 *Water-absorption test.*—Method 7031 of Federal Specification L–P–406 shall be employed for the water-absorption test. Three pieces, 1 by 2 inches, shall be cut from the shells of the hats prepared in acordance with 4.5.1 and after subjection to one of the strength tests specified in 4.5.3, 4.5.4, or 4.5.5. Water absorption reported shall be the average of that determined for the three specimens.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified, commercial packages are acceptable.

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5.2 Packing.—Unless otherwise specified, shipping containers for the subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.—Unless otherwise specified, shipping containers shall be marked with the name of the commodity, size, and quantity contained therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 Ordering data.—Procurement documents should specify the following and should exercise any desired options offered herein (see 3.3.4, 3.4.4, 5.1, 5.2, 5.3, and sec. 7).

(*a*) Title, number, and date of this specification.

(*b*) Type and size required (see 1.2.1 and 1.2.2).

(*c*) Whether headbands should be adjustable or nonadjustable (see 3.4.2).

6.1.1 When headband assemblies (for replacement purposes) are ordered, state the manufacturer's type or model number of the helmet for which the replacement headband is ordered, or if possible, a representative helmet should be furnished in order that the manufacturer may have the necessary identifying information.

6.2 In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Federal Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Navy Department, Bureau of Ships, Washington 25, D. C., and arrange to have the products that they propose to offer to the Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Navy Department, Bureau of Ships, Washington 25, D. C.

6.3 Standard head forms.—The standard head forms mentioned throughout this specification are the model heads known as "A. M. L. Head Size Standards", a set of which is located at each of the following places:

115 Industrial Building
National Bureau of Standards
Washington, D. C.

Aero Medical Laboratory
Wright Air Development Center
Wright-Patterson Air Force Base, Ohio

Material Laboratory
New York Naval Shipyard
Brooklyn 1, New York

6.4 Transportation description.—Transportation description applicable to these items is:

Helmets, not millinery, hard fiber, waterproofed, in boxes: Carload weight 12,000 pounds subject to Rule 34, Consolidated Freight Classification.

Helmets, not millinery, hard fiber:

Nested (preferred method of shipment to obtain lowest transportation charge), in boxes: Truckload weight, 6,000 pounds.

Not nested, in boxes: Truckload weight, 4,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standards, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

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7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standards.*

Federal Specifications:

NN-B-621—Boxes, Wood, Nailed and Lock-Corner.

UU-P-268—Paper, Kraft, Wrapping.

LLL-B-631—Boxes, Fiber, Corrugated (for Domestic Shipment).

Military Specifications:

JAN-P-106—Packaging and Packing for Overseas Shipment; Boxes, Wood, Nailed.

MIL-L-10547—Liners, Case, Waterproof.

Military Standard:

MIL-STD-129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Packaging.*—Each helmet shall be wrapped in a sheet of kraft wrapping paper having a minimum basis weight of 20 pounds per ream (24 by 36 inches) and conforming to grade B of Federal Specification UU-P-268. The paper shall be molded over the crown of the helmet, and extend around the brim and shall be tucked on the inside of the crown. Five wrapped helmets shall be stacked together by nesting the crowns one within the other.

7.2.3 *Packing.*

7.2.3.1 *For domestic shipment and storage.*—Unless otherwise specified, 20 helmets packaged as specified in 7.2.2. shall be packed in fiberboard or nailed wood boxes as specified in 7.2.3.1.1 and 7.2.3.2.1.

7.2.3.1.1 Fiberboard boxes shall conform to style RSC of Federal Specification LLL-B-631, except that fiberboard shall have a minimum bursting strength of 200 p. s. i. (Mullen or Cady).

7.2.3.1.2 Nailed wood boxes shall be constructed and closed in accordance with style 2 or 4 of Federal Specification NN-B-621.

7.2.3.2 *For overseas shipment.*—Unless otherwise specified, 20 helmets, packaged as specified in 7.2.2, shall be packed in a nailed box, constructed, closed, and strapped in accordance with style 2 or 4 of Military Specification JAN-P-106. The containers shall be fitted with a waterproof case liner conforming to type I, grade A or B, class 2, of Military Specification MIL-L-10547 with seams and closures sealed in accordance with the appendix thereto.

7.2.3.3 *For domestic and overseas shipment.*—Each fiberboard or nailed wood box shall be provided with a full-height slotted partition made of double-faced corrugated fiberboard, as specified for the fiberboard box, which shall divide the box into four equal sections. The shipping container and innerpacking shall be designed so that each stack of five helmets shall fit snugly in a partitioned section of the box.

7.2.3.3.1 *Marking.*—In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Military Standard MIL-STD-129. The nomenclature shall be as follows:

Helmet, Construction Worker's.

MILITARY INTEREST:

Navy—Sh S Y

Air Force.

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

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GGG-H-00142a(NAVY SHIPS)
29 OCTOBER 1952

SUPERSEDING
GGG-H-142(NAVY SHIPS)
8 OCTOBER 1951

INTERIM FEDERAL SPECIFICATION

HATS, PROTECTIVE

This interim Federal specification has been developed by the Navy Department, Bureau of Ships, based upon currently available technical information. It has not been approved for promulgation as a Federal specification and is subject to modification. This specification is in interim form only, and, pending its promulgation by the Administrator, General Services, as a regularly coordinated Federal specification, it may be used in procurement.

1. SCOPE AND CLASSIFICATION

1.1 Scope. - This specification covers protective hats for protection of the wearer's head against impacts of objects and contacts with exposed electrical conductors. The hats are intended for protection against these hazards in mining, building construction, tunnelling, timber work, manufacturing, etc.

1.2 Classification. - The hats shall be furnished in the following types and sizes specified in the contract or order.

Type I - Full brim.
Type II - Brimless with visor.

1.3 Sizes. - The hats shall be furnished in the head sizes specified in the contract or order. The hats with adjustable headbands shall have the range of sizes indicated on an instruction sheet or on a label pasted to the inside top of the crown of the shell (see 3.8).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
L-P-406 - Plastics, Organic: General Specifications, Test Methods.
KK-L-136 - Leather; Artificial (Upholstery).

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C., or to the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Seattle. Prices may be obtained from the Superintendent of Documents or from the GSA Regional Offices.)

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Specifications and standards applicable to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Qualification. - Protective hats furnished under this specification shall be a product which has been tested and has passed the qualification tests specified in section 4 (see 6.2).

3.2 Component parts. - Each hat shall consist of a shell, headband, crown straps forming a cradle or hammock for supporting the hat on the wearer's head, and if specified in the contract or order, an adjustable chin strap and a winter lining.

3.3 Material. - All materials coming into contact with the skin shall be nonirritating.

3.3.1 Shell. - The shell shall be of smooth hard material which shall conform to the requirements as hereinafter specified.

3.3.2 Headband. - Unless otherwise specified, the headband shall be genuine vegetable tanned leather, full grained and soft. The headband shall be smoothly finished on the surface which will be in contact with the head. If specified in the contract or order, headband may be made of artificial leather conforming to type III, synthetic resin coated, of Specification KK-L-136.

3.3.3 Crown straps. - Crown straps shall be of webbing, closely woven.

3.3.4 Accessory materials. - If chin straps and winter linings are specified in the contract or order, they shall be made of the following materials:

3.3.4.1 Chin strap. - The adjustable chin strap shall be closely woven cotton or linen webbing or of genuine leather.

3.3.4.2 Winter lining. - The winter lining, shall consist of two layers of closely woven fast-dyed fabric. The outer layer shall be a water-repellent material and the inside layer, nonwater-repellent, plain-woven cotton flannel with the nap on the inside surface.

3.4 Design and construction. -

3.4.1 Shell. - The shell shall be of one-piece, seamless construction, dome shaped, with a full brim for type I hats and with a visor for type II hats.

3.4.1.1 Dome. - The dome shall be designed to provide the inside clearances as specified in 3.4.3.

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3.4.1.2 Brim, type I. - The brim for type I hats shall be an integral extension of the shell dome, and shall be continuous around the hat. The width of the brim shall be not less than 1-1/2 inches and not more than 3 inches measured from the inside edge of the shell, except that the width at the front may taper to not less than 1-1/4 inches. With the hat held in a horizontal position, the brim shall slope downward in a straight line at an angle of not less than 15 degrees nor more than 37 degrees from the horizontal.

3.4.1.3 Visor, type II. - The visor for type II hats shall extend forward from the crown not less than 1-1/2 inches, and shall be not less than 5-1/2 inches wide at the peak.

3.4.2 Headbands. - The headbands shall be adjustable or nonadjustable as specified in the contract or order. If adjustable, they shall be suitable for the range of head sizes specified (see 1.3 and 6.1). If nonadjustable, they shall be furnished in the specified head sizes. The headband shall conform in size to the marking of the hat or shall permit adjustment to the marked size or sizes. If made of leather, the thickness, determined at 10 points uniformly spaced throughout the circumference of the headband, shall be not less than 2/64 inch, when measured with a gage having a 5/16-inch-diameter contact foot under a 1-pound load. The surface of the headband in contact with the head shall be not less than 1-1/4 inches in width. The headband shall be securely attached either to the hat shell or to the crown straps for assuring its proper position in the hat. The manner of the attachment should not interfere with the adjustments of the crown straps. Padding or stiffener strips, if used as a regular part of the headband, shall be secured to the leather.

3.4.3 Crown straps. - Crown straps shall be not less than 1/2 inch in width. These straps when adjustably and properly interlaced shall form a cradle or hammock for supporting the hat on the wearer's head, for holding the hat away from the head and thus acting as a shock absorber when the hat is struck. The number of straps shall be adequate and their location in the hat shall be suitable for the above purpose. The straps may have one end attached with the loose end having a sewn loop, or may have both ends attached forming a bent loop. Through these loops a strong cord or lace shall pass for adjusting the cradle to suit the wearer's head leaving a clearance of not less than 1-1/2 inches between the top of the wearer's head and the underside of the top of the crown.

3.4.4 Accessories. - If specified in the contract or order, the following accessories shall be furnished with the hat:

3.4.4.1 Chin strap. - The adjustable chin strap shall be of not more than two pieces of webbing or leather of not less than 1/2 inch in width and not less than 16 inches in adjusted length excluding the attachments to the hat. The means for adjusting the chin strap shall assure a secure hold of the hat on the wearer's head and quick removal of the hat by releasing the strap. The chin strap shall also be adjustable for wearing at the back of the head. The webbing shall have no frayed or loose edges that may unravel. The leather, if used in the chin strap, shall be of suitable thickness, full grained, and smoothly finished on the surface that will be in contact with the chin. All metal parts, including rivets, shall be smooth and free from sharp or rough edges or projections. Rivet heads shall be round.

3.4.4.2 Winter lining. - The winter lining shall be designed to cover the skull, neck, and ears. The skull cap, part of the winter lining, shall be attached to the hat cradle or to the hat shell in such a manner that it can be readily removed and reattached. The earlug and neck part may be made either in one piece with the skull cap or attachable to it by tabs or a zipper. The neck and earlug shall be drawn close to the face by a lace or a chin strap.

3.5 Physical requirements. -

3.5.1 Insulation resistance. - The hat shall not show a leakage current in excess of 3 milliamperes when tested as specified in 4.5.2.

3.5.2 Impact resistance. - When hats are tested in accordance with 4.5.3, the average diameter of the impressions shall not exceed 4.5 millimeters. No individual specimen shall show a diameter of impression greater than 4.9 millimeters. (The diameters of impressions specified correspond, when using 2S-0 aluminum, to transmitted forces of approximately 850 and 1,000 pounds, respectively.)

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3.5.3 Penetration strength. - The hat shall neither be dented nor pierced for more than 3/8 inch, nor shall the hat be forced down over the standard head form, nor shall the straps pull out or break when tested as specified in 4.5.4.

3.5.4 Edge strength, type II. - When tested in accordance with 4.5.5, type II hats shall show a deflection under a 40-pound load of not more than 3/4 inch and an ultimate strength of not less than 50 pounds.

3.5.5 Flammability. - The thinnest section of the shell shall not burn at a rate greater than 3 inches per minute when tested as specified in 4.5.6.

3.5.6 Water absorption. - The hat shell material shall absorb not more than 5 percent of water when tested as specified in 4.5.7.

3.5.7 Weight. - The weight of each complete hat without winter lining and chin strap, shall not exceed 14-1/2 ounces.

3.5.8 Ventilation. - Provision shall be made between the shell and the headband, or in the headband proper, for a free passage of air for ventilation purposes.

3.6 Interchangeability. - The headband and crown-strap assembly shall be easily removable and replaceable, and shall be interchangeable among shells of the same size furnished by the same manufacturer, provided that a single type of lining is furnished. Where two types of linings are involved, neither lining need be usable as a replacement in shells originally provided with the other type.

3.7 Head size. - The head size or range of sizes to which the headband is adjustable shall be marked on labels cemented to or embossed upon the hat shell and sewn or cemented to the replaceable crown strap assembly.

3.8 Instruction sheet. - An instruction sheet which explains the proper method of wearing and adjusting the hat shall be furnished with each hat. The sheet shall be stowed in the hat so as to remain where placed until intentionally removed.

3.9 Marking. - Each hat shell shall be marked in a permanently legible manner with the manufacturer's name or trade-mark of such known character that the source of manufacture may be readily determined.

3.10 Workmanship. - The workmanship shall be first class in every respect. The protective hats shall have no sharp edges or burrs, and shall be free from any other defects which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Qualification tests at a Government laboratory. - Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.5.

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4.2 Sampling for lot acceptance. -

4.2.1 Lot. - All hats of each type presented for delivery at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.2.2 Sampling for inspection. - A random sample of hats shall be selected as specified in table I from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on sampling inspection requirements in accordance with Standard MIL-STD-105.

Table I - Sampling for visual and dimensional inspection
AQL (approx.) = 2.5 percent defective.

Number of hats in inspection lot	Number of hats in sample	Acceptance number (defectives)	Rejection number (defectives)
40 or under	10	0	1
41 to 110	15	1	2
111 to 300	25	1	2
301 to 500	35	2	3
501 to 800	50	3	4
801 to 1,300	75	4	5
1,301 to 3,200	110	6	7
3,201 and over	150	8	9

4.2.3 Sampling for lot acceptance tests. - Samples shall be taken at random from each inspection lot by the Government inspector for lot acceptance tests in accordance with table II.

Table II - Sampling for lot acceptance tests.

Number of hats in lot	Number of hats in sample
15 or under	None
16 to 40	3
41 to 110	6
111 to 300	12
301 to 800	18
801 to 3200	24
3201 and over	36

4.3 Inspection and tests. -

4.3.1 Visual and dimensional inspection. - Each of the sample hats selected in accordance with table I shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Any hat in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective hats in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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4.3.2 Lot acceptance tests. - Samples selected in accordance with 4.2.3 shall be subjected to the tests specified in 4.5.1 through 4.5.5.4 except that the impact test of 4.5.3 shall be conducted at the prevailing atmospheric temperature in lieu of the temperatures specified in 4.5.3.3. If any sample hat fails one or more of these tests the entire lot shall be rejected.

4.4 Rejected lots. - Rejected lots may be offered again for Government inspection provided the contractor has removed all nonconforming hats. The Government inspector shall again select and examine samples from such resubmitted lots to verify compliance with this specification.

4.5 Test procedures. -

4.5.1 Preparation of samples. - The sample hats shall be selected as specified in table II. Where it is evident that the sample hats under test have a protective coating over the basic material, the entire exterior surface of the shell shall be abraided until the basic material is exposed using No. 60 grit Garnet paper. All samples shall then be conditioned in an oven for 96 hours at 50° + or -2°C., then cooled in a desiccator and exposed for 96 hours in an atmosphere having 50 + 2 percent relative humidity and a temperature of 25° + 2°C. Immediately following conditioning, 1/3 of the samples shall be subjected to the test specified in 4.5.3, 1/3 to the test specified in 4.5.4, and the remainder to the test specified in 4.5.5.

4.5.2 Insulation resistance test. -

4.5.2.1 Test specimens. - All of the hats in one of the groups specified in 4.5.1 prior to subjection to the impact test or to the penetration test shall be subjected to the insulation resistance test.

4.5.2.2 Apparatus. - The test apparatus shall consist of the following:

(a) A vessel containing a solution of 1 ounce of sodium chloride per gallon of water and of sufficient size to submerge an inverted hat to within 1/2 inch of the junction of brim with the dome.
(b) Wire frame for suspending inverted hat in sodium chloride solution.
(c) Source of 60-cycle alternating current with 2,200 volts (r. m. s.)
(d) Wiring and terminals for application of voltage across dome of hat.
(e) Voltmeter of sufficient capacity.
(f) Milliammeter of sufficient capacity and accuracy to measure the specified current.

4.5.2.3 Procedure. - An inverted hat with the head-harness assembly removed shall be placed on the wire frame and submerged in the solution to within 1/2 inch of the junction of brim with the dome. The hat shall also contain the solution to the same level. If the hat is provided with holes in the crown near the brim, the hat shall be submerged to within 1/2 inch of the holes. One terminal from the current source shall be connected to the wire frame which supports the hat and the other shall be immersed in the sodium chloride solution which is contained by the hat. The 2,200 volts shall then be applied for 1 minute. If passage of current is evidenced by deflection of the voltmeter, the hat shall be considered as having failed to conform to 3.5.1. If no deflection of voltmeter occurs, a milliammeter is placed in the circuit and a reading taken.

4.5.2.4 Report. - For each specimen, the leakage current or evidence of complete breakdown shall be reported.

4.5.3 Impact test. -

4.5.3.1 Test specimens. - All hats in the first group prepared in accordance with 4.5.1 shall be subjected to the impact test.

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4.5.3.2 Apparatus. - The test apparatus shall consist of the following:

(a) A standard head form (see 6.3) of the same headsize as the hat under test. The standard head forms are of 4 sizes: small, medium, large and extra large, representing head sizes of 6-3/4, 7, 7-1/4 and 7-1/2, respectively. Test block forms which simulate the standard head forms may be employed. The blocks shall duplicate the crown dimensions and contours from the top down to the ear and eye levels, including the top portion of the ears. Each test block shall have a hole drilled vertically from the top of the crown to facilitate measurement of hat crown clearance.
(b) A Brinell hardness penetrator assembly as shown on figure 1. The impression bar shall be of 2S-0 aluminum not less than 1/4 inch thick and having a Brinell hardness of 20 to 25 as measured with a 500-kilogram load using a 10-millimeter ball. The Brinell penetrator shall be a hardened steel ball 1/2 inch in diameter.
(c) An 8-pound spherical weight, approximately 3.8 inches in diameter.
(d) A Brinell microscope or other micrometer microscope accurate to 0.1 millimeter.

4.5.3.3 Test conditions. - Impact tests shall be conducted with the specimens maintained at temperatures of 20° ± 2°F. and 120° ± 2°F. Specimens shall be subjected to the test temperature for at least 4 hours prior to the impact test.

4.5.3.4 Procedure. - The specimens specified in 4.5.3.1 shall be mounted on the appropriate standard head form in a position similar to that when worn on a man's head. For purposes of testing, headbands of sizes intermediate to the standard form size shall be padded with felt to reduce the headband to fit the nearest applicable form size, for example a size 7-1/8 hat shall be padded to fit a size 7 form. The hats shall be adjusted to a clearance of 1-1/4 inches between crown of test block and the underside of the top of hat crown by placing a 25-pound weight on the hat and measuring by means of a rod inserted in the hole drilled in test block (see 4.5.3.2 (a)). The hat and block assembly shall then be mounted on the Brinell hardness penetrator assembly. The 8-pound spherical weight shall then be dropped vertically on the center of the crown from a height of 5 feet, measured from the bottom of the ball to the top of the hat. The diameter of the impression, produced on the impression plate as a result of the impact, is then measured to the nearest 0.1 millimeter with the Brinell or other suitable micrometer microscope.

4.5.3.5 Report. - The average diameter of impression of all hats tested under one group and the greatest individual diameter of impression from the same group shall be reported.

4.5.4 Penetration test. -

4.5.4.1 Test specimens. - All hats in the second sample group prepared in accordance with 4.5.1 shall be subjected to the penetration test.

4.5.4.2 Apparatus. - The test apparatus shall consist of standard head forms or test blocks as specified in 4.5.3.2 (a) and a 16-ounce plumb bob with a steel point having an included angle of 36 degrees.

4.5.4.3 Procedure. - The specimen shall be mounted on a test block as specified in 4.5.3.4 and the plumb bob dropped squarely onto the center of the crown from a height of 10 feet, measured from the bottom point of the plumb bob to the top of the hat. The depth the plumb bob has pierced the hat shall be measured to the nearest 1/32 inch and shall include the thickness of the shell material.

4.5.4.4 Report. - For each specimen, the depth of penetration to the nearest 1/32 inch shall be reported.

4.5.5 Edge strength test, type II. -

4.5.5.1 Test specimens. - All hats in the third group prepared in accordance with 4.5.1 shall be subjected to the edge strength test.

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4.5.5.2 Apparatus. - The test apparatus shall consist of the following:

(a) A universal testing machine capable of applying the loading specified in 3.5.4 at a rate of head travel of 0.05 ± 0.01 inch per minute.
(b) A dial gage for measuring deflections under loading and accurate to 0.01 inch.
(c) A jig designed to support a type II protective hat on its side on the bed plate of the testing machine so that a plane intersecting the edge of the crown is vertical and the long axis of the hat horizontal.

4.5.5.3 Procedure. - With a specimen mounted in the jig, the traveling head of the test machine shall be brought into contact with the edge of crown and the rate of deflection adjusted to 0.05 ± 0.01 inch per minute. The dial gage shall be adjusted to measure the travel of the moving head. Deflections shall be measured at intervals of loading of 10 pounds. Loading shall be continued to failure.

4.5.5.4 Report. - For each specimen, the deflection under a load of 40 pounds and the ultimate load at failure shall be reported.

4.5.6 Flammability test. -

4.5.6.1 Test specimens. - All hats in one of the three groups prepared in accordance with 4.5.1, and after subjection to one of the strength tests specified in 4.5.3, 4.5.4 or 4.5.5, shall be subjected to the flammability test.

4.5.6.2 Apparatus. - The test apparatus shall consist of the following:

(a) Rigid support and clamps for mounting test strips in specified position.
(b) Alcohol lamp or gas burner.
(c) Stop watch.

4.5.6.3 Procedure. - Three strips approximately 1/2 inch wide and 5 inches long shall be cut from the brim or sides of the crown, whichever is thinner. These strips shall be cut from a portion of the hat so as to give as flat a section as practicable. Lines shall be drawn across each strip at 1/2-inch intervals, measured along the surface and starting from one end. Clamp the other end to the rigid support so that the longitudinal axis is horizontal and the transverse axis is inclined 45 degrees. The alcohol lamp or gas burner, with a neutral, blue flame 1/2 to 3/4 inch in height, shall be placed under the free end of the strips, so adjusted that the flame tip is just in contact with the material. A stop watch shall be started when the flame is placed in contact with the strip under test. At the end of 30 seconds the flame shall be removed and the strip allowed to burn noting the rate of burning.

4.5.6.4 Report. - The rate of burning shall be reported as the average for the three specimens.

4.5.7 Water absorption test. - Method 7031 of Specification L-P-406 shall be employed for the water absorption test. Three pieces, 1 by 2 inches, shall be cut from the shells of the hats prepared in accordance with 4.5.1 and after subjection to one of the strength tests specified in 4.5.3, 4.5.4, or 4.5.5. Water absorption reported shall be the average of that determined for the three specimens.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Unless otherwise specified, commercial packages are acceptable.

5.2 Packing. - Shipping containers for the subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking. - Unless otherwise specified, shipping containers shall be marked with the name of the commodity, size, and quantity contained therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2 and 1.3).
(c) Whether headbands should be genuine or artificial leather (see 3.3.2).
(d) What, if any, accessories should be furnished (see 3.3.4).
(e) Whether headbands should be adjustable or nonadjustable (see 3.4.2).

6.1.1 When headband assemblies (for replacement purposes) are ordered, state the manufacturer's type or model number of the hat for which the replacement headband is ordered, or if possible, a representative hat should be furnished in order that the manufacturer may have the necessary identifying information.

6.2 In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Federal Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Navy Department, Bureau of Ships, Washington 25, D.C., and arrange to have the products that they propose to offer to the Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Navy Department, Bureau of Ships, Washington 25, D.C.

6.3 Standard head forms. - The standard head forms mentioned throughout this specification are the model heads known as "A.M.L. Head Size Standards", a set of which are located at each of the following places:

115 Industrial Building
National Bureau of Standards
Washington, D.C.

Aero Medical Laboratory
Wright Air Development Center
Wright-Patterson Air Force Base, Ohio

Material Laboratory
New York Naval Shipyard
Brooklyn 1, New York

6.4 Transportation description. - Transportation description applicable to these items is:

Hats, not millinery, hard fibre, waterproofed.
In boxes.
Carload weight 12,000 pounds subject to Rule 34, Consolidated Freight Classification.
Hats, not millinery, hard fibre
Nested (Preferred method of shipment to obtain lowest transportation charge).
In boxes.
Truckload weight, 6,000 pounds.
Not nested.
In boxes.
Truckload weight, 4,000 pounds

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Patent notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standards, of the issue in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force. -

7.2.1 Applicable specifications and standards. -

SPECIFICATIONS

FEDERAL
NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.
UU-P-268 - Paper, Kraft, Wrapping.
LLL-B-631 - Boxes, Fiberboard, Corrugated (for Domestic Shipment).

MILITARY
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed.
JAN-P-125 - Packaging and Packing for Overseas Shipment - Barrier-Materials, Waterproof, Flexible.
MIL-A-140 - Adhesive, Water-Resistant, Waterproof Barrier-Material.

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-129 - Marking of Shipments.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

7.2.3 Packaging. - Each hat shall be wrapped in a sheet of kraft wrapping paper having a minimum basis weight of 20 pounds per ream (24 by 36 inches) and conforming to grade B of Specification UU-P-268. The paper shall be molded over the crown of the hat, and extend around the brim and shall be tucked on the inside of the crown. Five wrapped hats shall be stacked together by nesting the crowns one within the other.

7.2.4 Packing. -

7.2.4.1 For domestic shipment. - Unless otherwise specified, 20 hats packaged as specified in 7.2.3 shall be packed in fiberboard or nailed wood boxes as specified in 7.2.4.1.1 and 7.2.4.1.2.

7.2.4.1.1 Fiberboard boxes shall conform to style RSC of Specification LLL-B-631, except that fiberboard shall have a minimum bursting strength of 200 p.s.i. (Mullen or cady).

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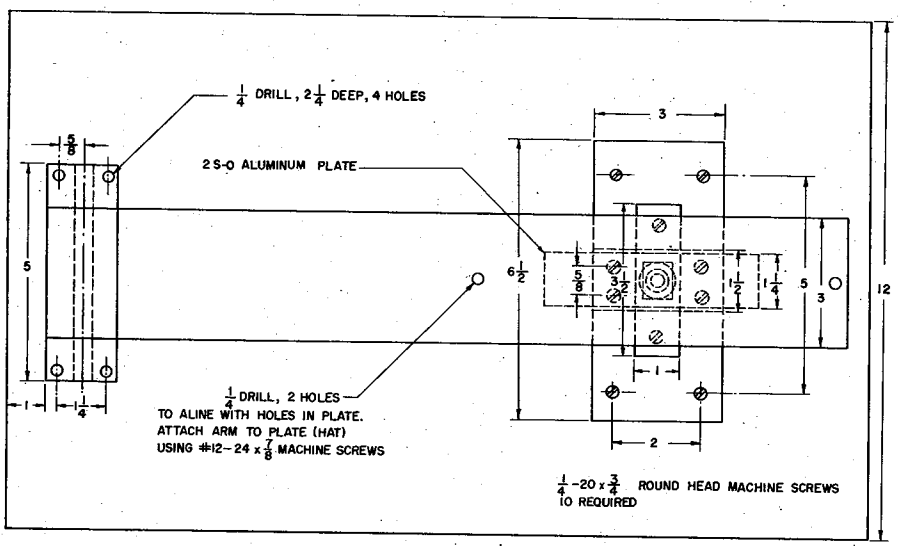
7.2.4.1.2 Nailed wood boxes shall be constructed and closed in accordance with style 2 or 4 of Specification NN-B-621.

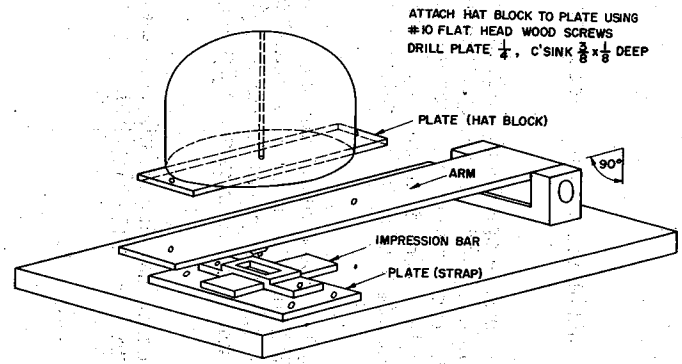
7.2.4.2 For overseas shipment.- Unless otherwise specified, 20 hats, packaged as specified in 7.2.3, shall be packed in a nailed box, constructed, closed, and strapped in accordance with style 2 or 4 of Specification JAN-P-106. Each box shall be provided with a sealed case liner made of water-proof barrier material conforming to style L2 or M of Specification JAN-P-125. Closures shall be completely sealed with a continuous seam at least 3/4 inch wide, of water-resistant adhesive conforming to Specification MIL-A-140.

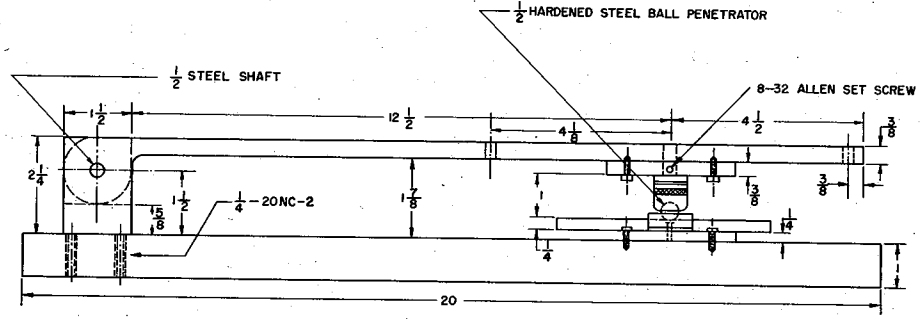
7.2.4.3 For domestic and overseas shipment.- Each fiberboard or nailed wood box shall be provided with a full-height slotted partition made of double-faced corrugated fiberboard as specified for the fiberboard box, which shall divide the box into four equal sections. The shipping container and innerpacking shall be designed so that each stack of five hats shall fit snugly in a partitioned section of the box.

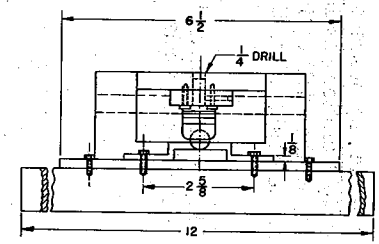
7.2.4.3.1 Marking.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL-STD-129.

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REMARKS

THE COMBINED THICKNESS OF IMPRESSION BAR AND PLATE (STRAP) SHALL EQUAL 1/2.

FIGURE 1.-BRINELL HARDNESS PENETRATOR ASSEMBLY.

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